

EE 102 Week 5, Lecture 1 (Fall 2026)

Date: September 28, 2026

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1 Goals

The main goal of this lecture is to learn how to visualize the process of convolution using graphs.

2 Review: Convolution definition

Recall that in continuous-time, the output $y(t)$ of an LTI system with input $x(t)$ and impulse response $h(t)$ is given by the convolution integral:

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau.$$

Pop Quiz 2.1: Check your understanding!

Prove that convolution is commutative, i.e., show that $x(t) * h(t) = h(t) * x(t)$.

Solution on page [14](#)

3 Discrete time convolution

Similar to the derivation for continuous-time convolution, we can derive the discrete-time convolution sum. Consider a discrete-time LTI system with input $x[n]$, output $y[n]$, and impulse response $h[n]$. Note that for a discrete-time impulse $\delta[n]$, the output is $h[n]$. Recall the sifting property of the discrete-time impulse:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n - k].$$

Using linearity and time-invariance of the system, we can write the output as

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k].$$

This is the discrete-time convolution sum, denoted by $y[n] = x[n] * h[n]$.

4 Example: A discrete-time echo system

An audio receiver system produces an echo. When excited by a unit impulse, it responds with an echo of magnitude 1 at $n = 0$ that decays exponentially as α^n for $\alpha \in (0, 1)$ until $n = 5$ (that is, for six samples in total). You may assume that $\alpha = \frac{1}{2}$ for numerical parts. Answer the following:

- (A) Sketch the impulse response $h[n]$ and label $h[0], h[1], \dots, h[5]$.
- (B) We want to understand the kind of echo that will be produced when the audio receiver system is excited by a pulse input of unit amplitude lasting three samples, starting at $n = 0$ and ending before $n = 3$ (so $x[n] = 1$ for $n = 0, 1, 2$). Find $y[n]$ for this input using convolution and show your steps.

The impulse response of the system is

$$h[n] = \begin{cases} \alpha^n, & 0 \leq n \leq 5, \\ 0, & \text{otherwise,} \end{cases}$$

The input is a unit amplitude tone that starts at $n = 0$ and lasts three samples. So, we can write the pulse signal for the input $x[n]$ as

$$x[n] = u[n] - u[n - 3] = \begin{cases} 1, & n = 0, 1, 2, \\ 0, & \text{otherwise.} \end{cases}$$

Now, we can compute the output $y[n]$ using convolution:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k],$$

and give $y[n]$ explicitly for all n where it is nonzero. It is important that we are careful about all values of n for which $y[n]$ is nonzero. Echoes can last longer than the original sound!

4.1 Convolution computation (without visualizing)

Let us compute the output for various values of n using the convolution sum directly:

n	$y[n]$
-2 :	0
-1 :	0
0 :	1
1 :	$1 + \alpha$
2 :	$1 + \alpha + \alpha^2$
3 :	$\alpha + \alpha^2 + \alpha^3$
4 :	$\alpha^2 + \alpha^3 + \alpha^4$
5 :	$\alpha^3 + \alpha^4 + \alpha^5$
6 :	$\alpha^4 + \alpha^5$
7 :	α^5
8 :	0
9 :	0
10 :	0

$$\Rightarrow \quad \text{with } \alpha = \frac{1}{2} : \quad y[0..7] = \left[1, \frac{3}{2}, \frac{7}{4}, \frac{7}{8}, \frac{7}{16}, \frac{7}{32}, \frac{3}{32}, \frac{1}{32} \right].$$

So, we find that the output $y[n]$ is nonzero for $n = 0, 1, \dots, 7$. In general, the length of the output from discrete-time convolution is $N + M - 1$ where N and M are the lengths of the two signals being convolved. Here, the length of $x[n]$ is 3 and the length of $h[n]$ is 6, so the length of $y[n]$ is $3 + 6 - 1 = 8$ samples. **This is important!**

4.2 Visualizing convolution (with graphs)

Now, we will solve this by using illustrations of convolution. For each index $n = 0, 1, \dots, 7$, draw three plots in a row for each n :

$$x[k], \quad h[n - k], \quad \text{and} \quad x[k]h[n - k],$$

all as functions of k . The output sample is the sum of the product samples:

$$y[n] = \sum_k x[k]h[n - k].$$

Let's start by drawing $h[n]$ for $\alpha = \frac{1}{2}$:

4.3 Idea: Flip 'h' and slide through 'x'

The sequence $x[k]$ does not change with n ; only $h[n - k]$ does.

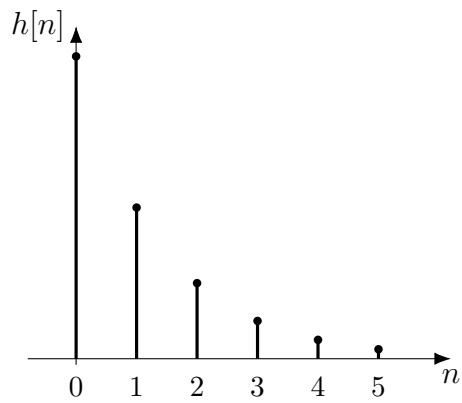


Figure 1: Impulse response $h[n]$ for $\alpha = \frac{1}{2}$.

For $n = 0$, the convolution is visualized in Figure 2.

Then, for $n = 1$, the convolution is visualized in Figure 3 and for all other nonzero values of n see Figures 4, 5, 6, 7, 8, and 9.

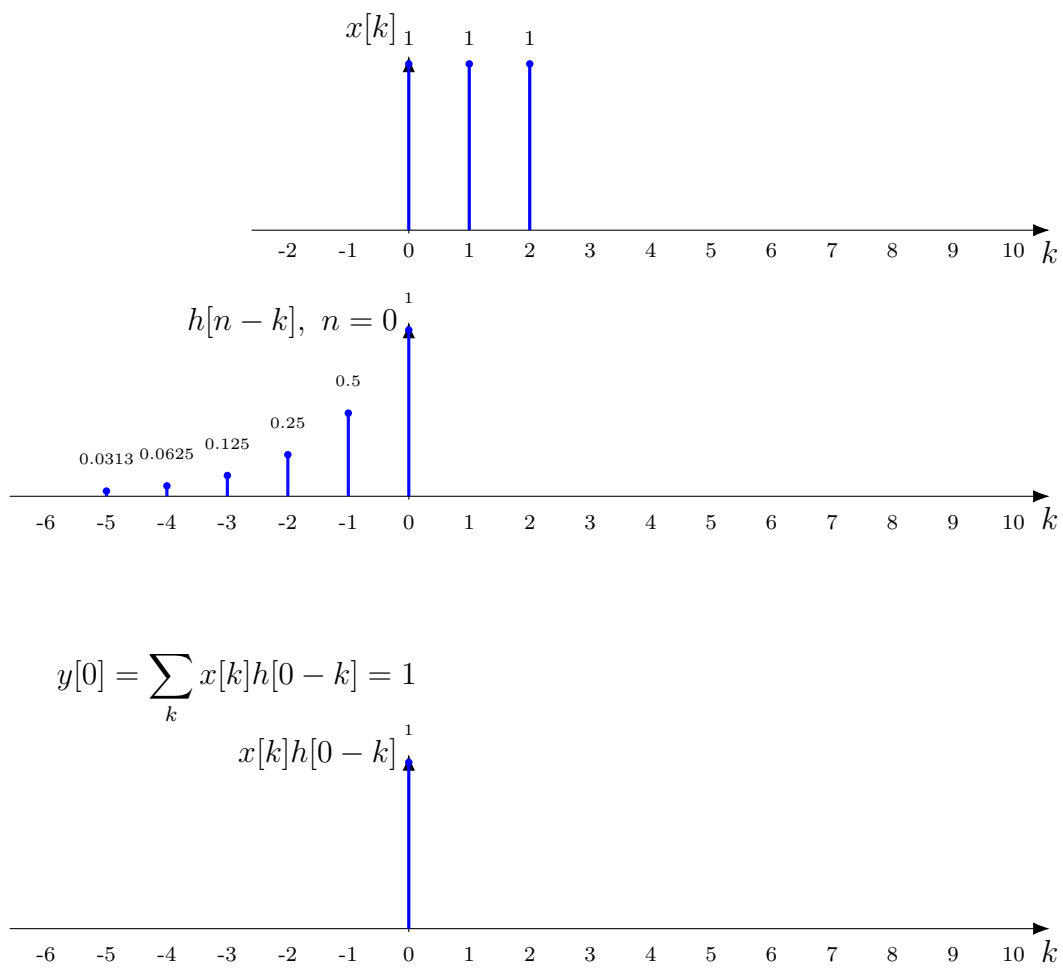


Figure 2: Convolution for $n = 0$.

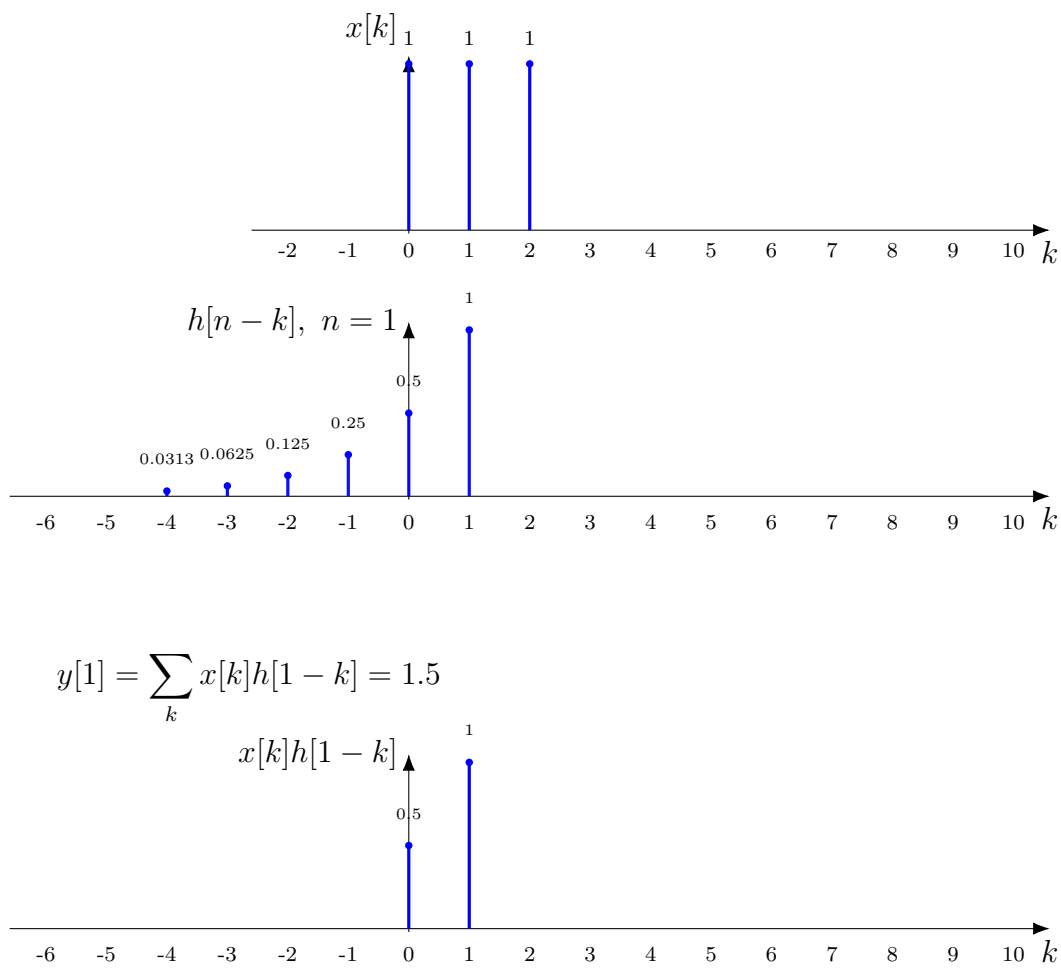


Figure 3: Convolution for $n = 1$.

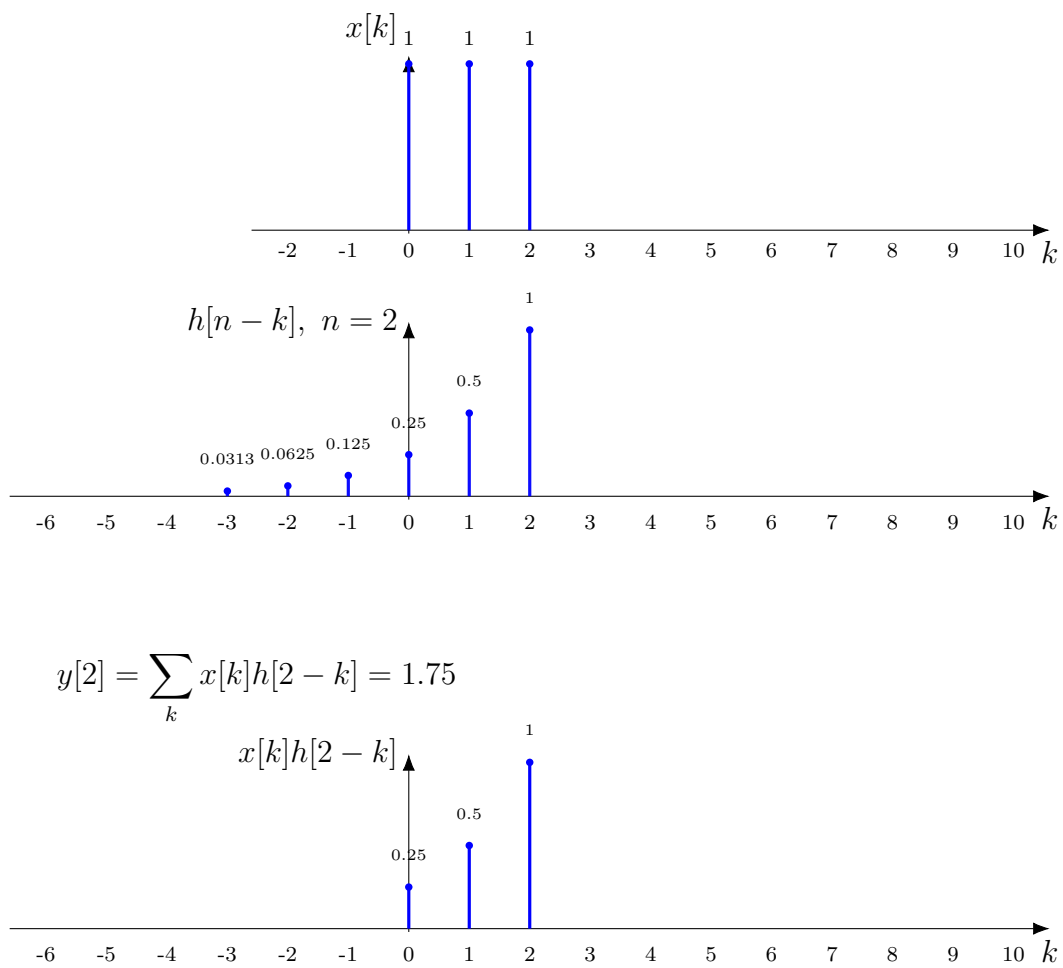


Figure 4: Convolution for $n = 2$.

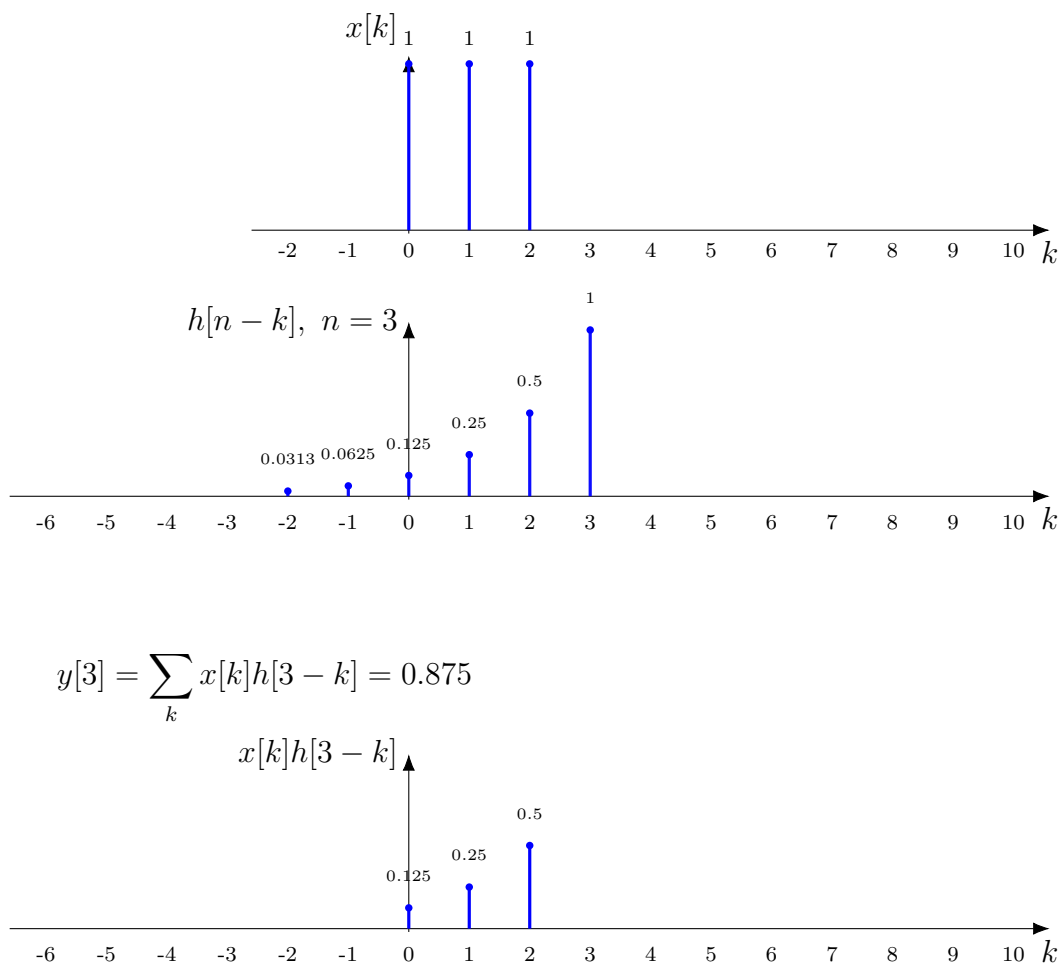


Figure 5: Convolution for $n=3$.

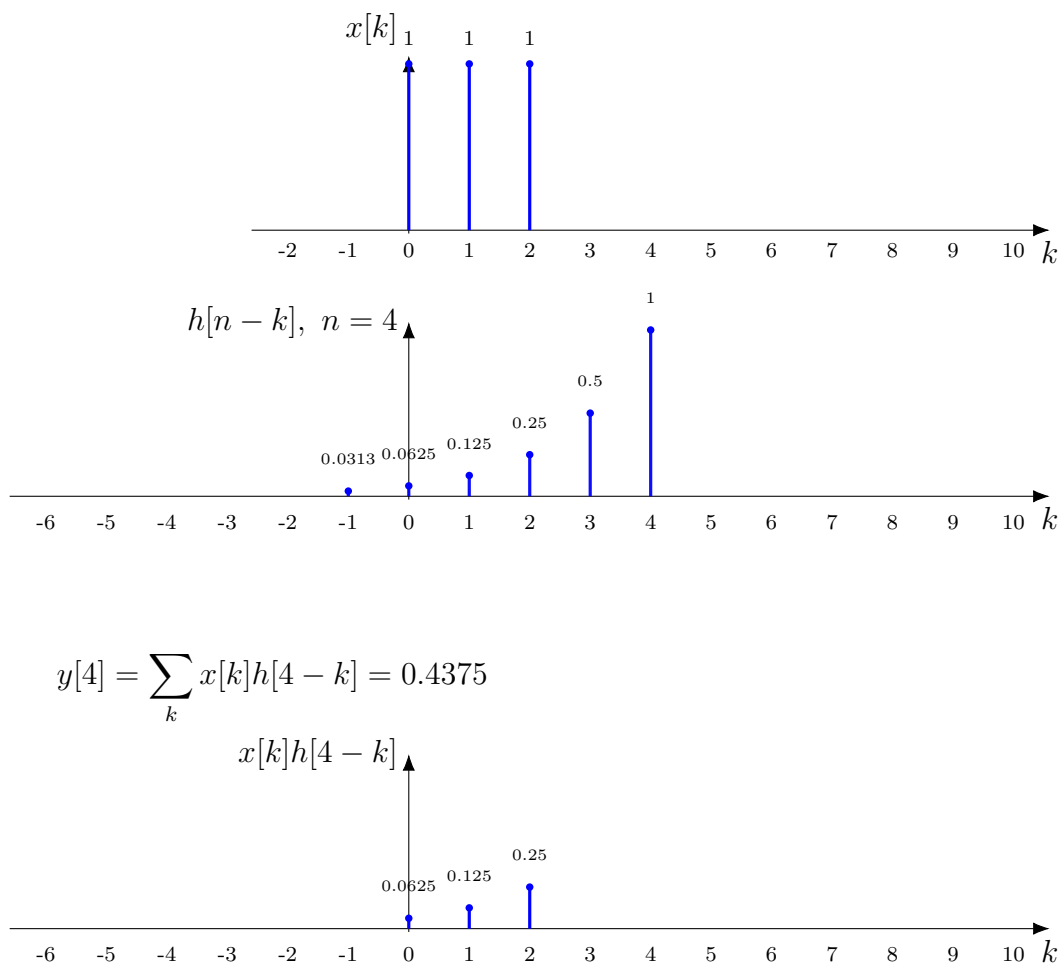


Figure 6: Convolution for $n = 4$.

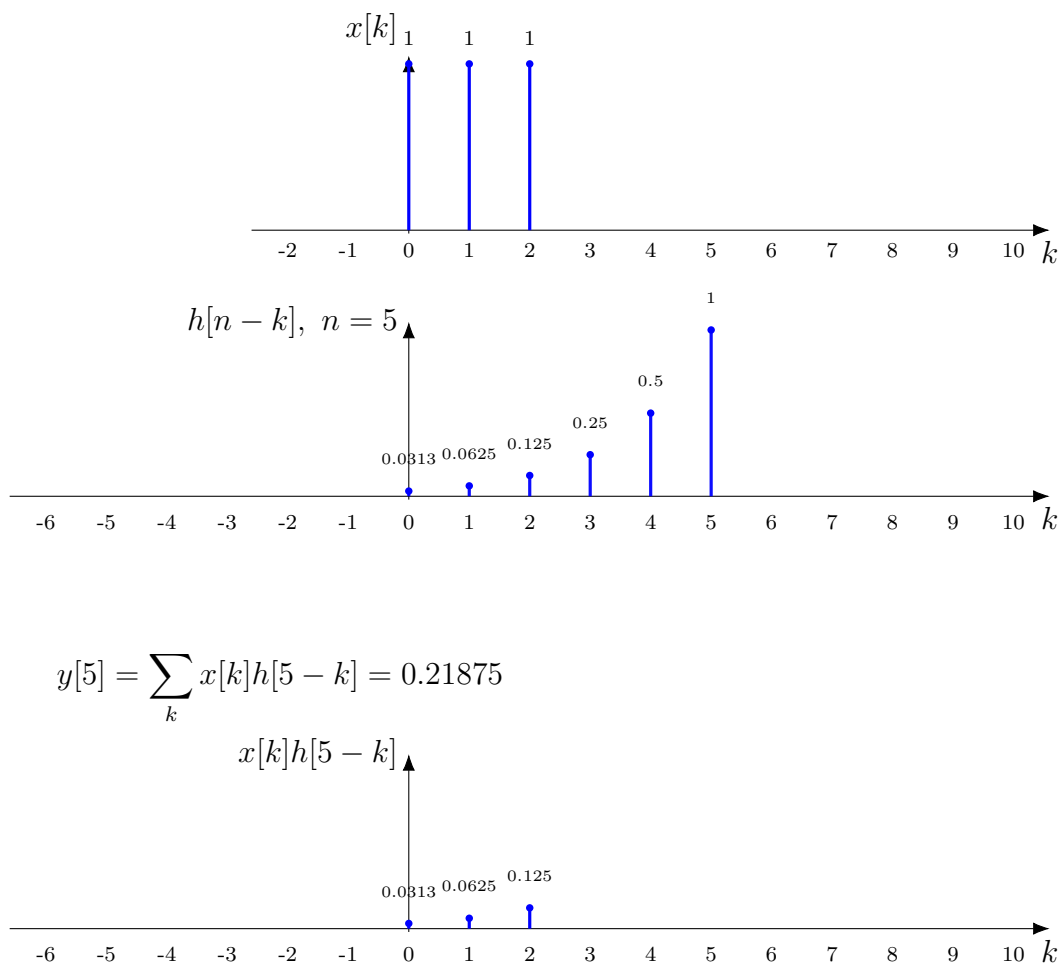


Figure 7: Convolution for $n = 5$.

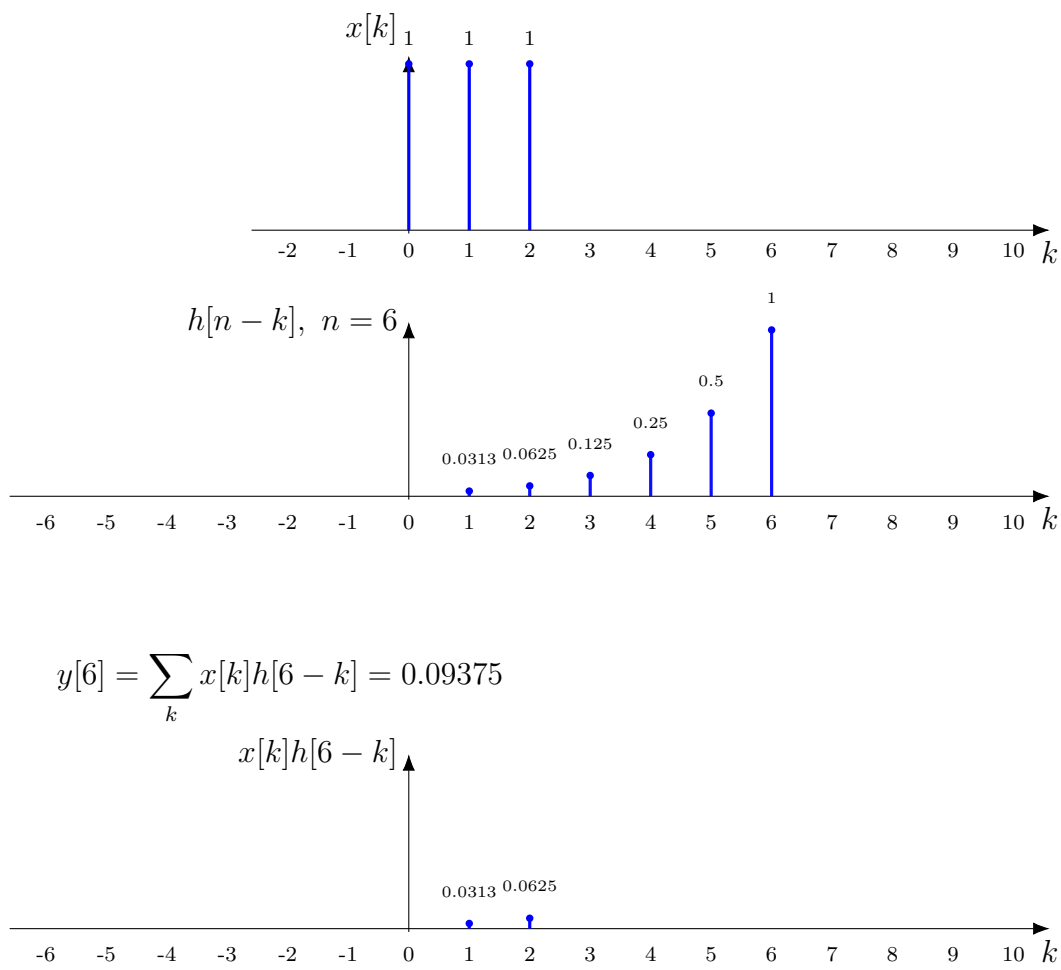


Figure 8: Convolution for $n = 6$.

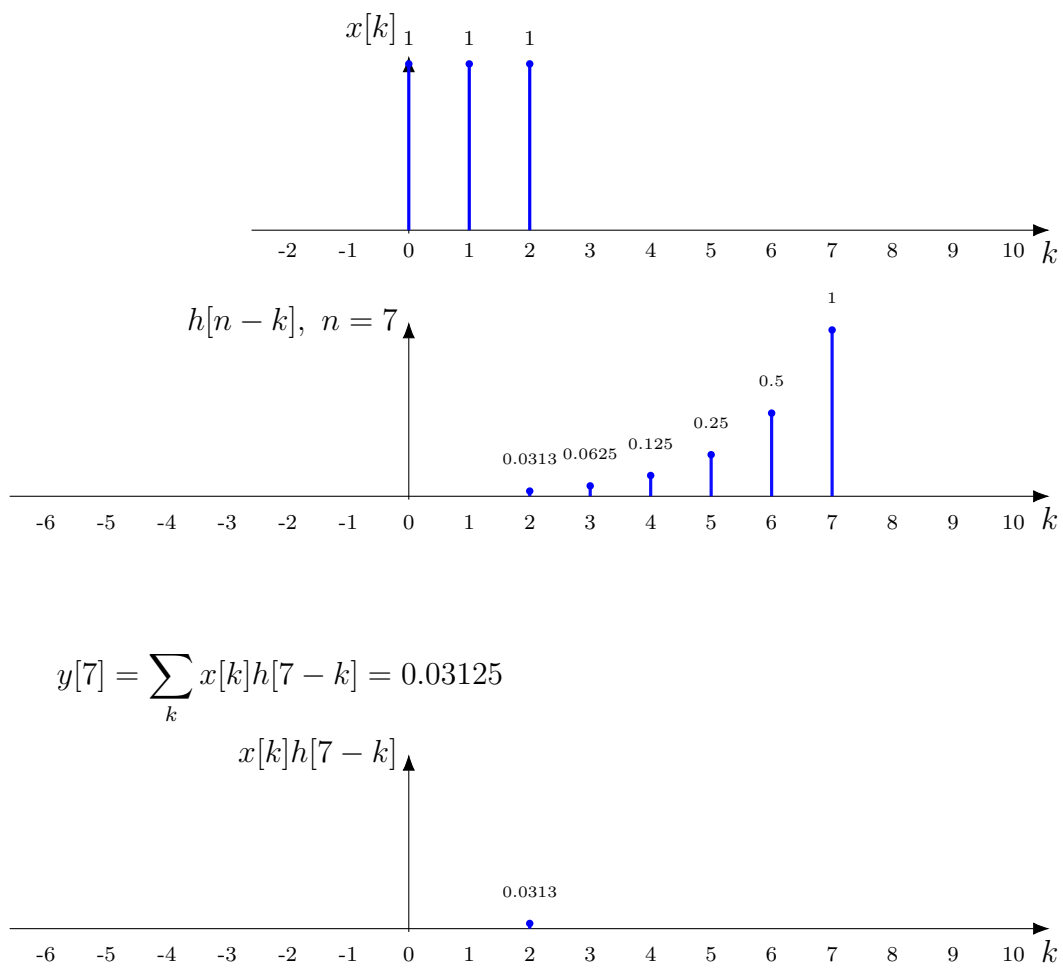


Figure 9: Convolution for $n = 7$.

Pop Quiz Solutions

Pop Quiz 2.1: Solution(s)

Write the convolution integral for left-hand side as

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau.$$

Now, let $s = t - \tau$. Then, $\tau = t - s$ and $d\tau = -ds$. When τ goes from $-\infty$ to ∞ , s goes from ∞ to $-\infty$. Thus, we can rewrite the integral as

$$x(t) * h(t) = \int_{\infty}^{-\infty} x(t - s)h(s)(-ds) = \int_{-\infty}^{\infty} h(s)x(t - s) ds.$$

This is exactly the convolution integral for $h(t) * x(t)$. Hence, convolution is commutative.